

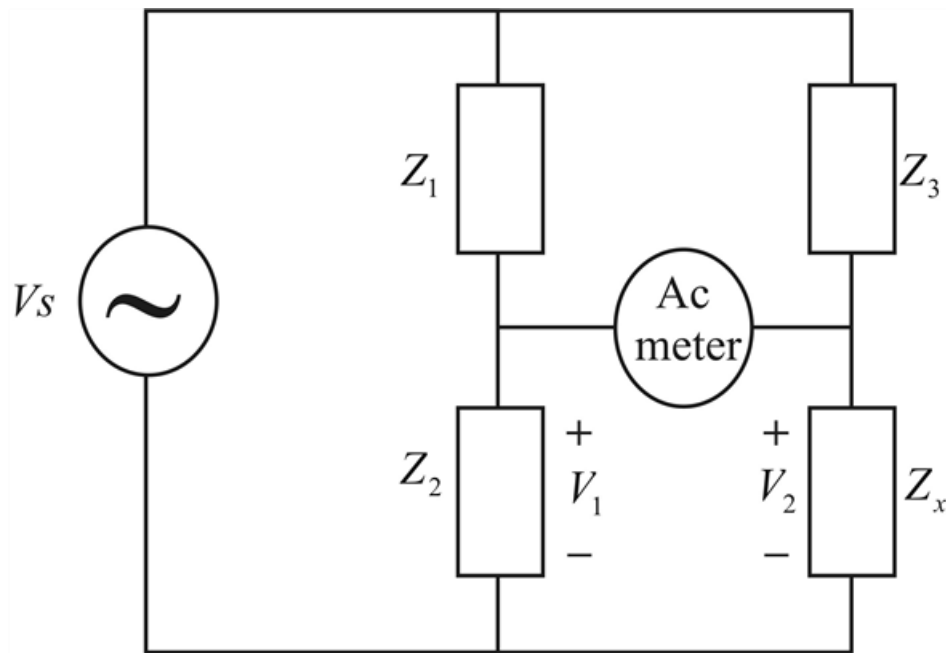
Problem

In the ac bridge circuit of Fig. 9.37, suppose that balance is achieved when Z_1 is a 4.8-k Ω resistor, Z_2 is a 10- Ω resistor in series with a 0.25- μ H inductor, Z_3 is a 12-k Ω resistor, and $f = 6$ MHz. Determine the series components that make up Z_x .

Step-by-step solution

Step 1 of 4

Given circuit is,



Step 2 of 4

The bridge is balanced when no current flows through the meter. This means that

$$V_1 = V_2$$

Given that

$$f = 6 \text{ MHz}$$

$$Z_1 = 4.38 \text{ k}\Omega$$

$$Z_2 = 10 + j \times 2\pi \times 6 \times 10^6 \times (0.25 \times 10^{-6})$$

$$Z_2 = 10 + j9.4248\Omega$$

$$Z_3 = 12 \text{ k}\Omega$$

Step 3 of 4

By using voltage division principle 6above circuit

$$V_1 = V_s \times \left[\frac{Z_2}{Z_2 + Z_1} \right]$$

And

$$V_2 = V_s \times \left[\frac{Z_x}{Z_x + Z_3} \right]$$

Therefore

$$V_s \times \left[\frac{Z_2}{Z_2 + Z_1} \right] = V_s \times \left[\frac{Z_x}{Z_x + Z_3} \right] (\because V_1 = V_2)$$

$$\frac{Z_2}{Z_2 + Z_1} = \frac{Z_x}{Z_x + Z_3}$$

$$Z_2 Z_x + Z_2 Z_3 = Z_2 Z_x + Z_1 Z_x$$

$$Z_2 Z_3 = Z_x (Z_2 + Z_1 - Z_2)$$

$$Z_2 Z_3 = Z_1 Z_x$$

$$\therefore Z_x = \frac{Z_2 Z_3}{Z_1}$$

Step 4 of 4

Substitute Z_1, Z_2 and Z_3 in the above equation

$$Z_x = \frac{(10 + j9.4248)(12 \times 10^3)}{4.8 \times 10^3}$$

$$\therefore Z_x = 25 + j23.562 \Omega$$

$$Z_x = 25 + j \times 2\pi \times 6 \times 10^6 \times (0.625 \times 10^{-6})$$

Therefore,

Z_x is a 25Ω resistor in series with a $0.625 \mu\text{H}$ inductor.